

Outline

Choose a standardized graduate level test

- Study how what it's like, what current applications there are to study
 - Pick something I want to change based on learning science

Build within Anki Codebase

- Ship a desktop and mobile app
 - Mobile app can be inside emulator
 - IOS(swift)

AI Features

- Brainlift should help to decide what features
 - Should have an eval harness
1. Confirm Tech Stack using a mermaid diagram
 2. Run Anki Locally
 3. Confirm PRD
 4. Spiky POV into features

Brainlift

Brainlift - MCAT Study Redesign

DOK 3

DOK 2

Category 1: Question Banks build pattern recognition, not understanding

- Sources: Larsen et al. (2009, Medical Education) · Schmidmaier et al. (2011, Advances in Health Sciences Education)

DOK 2 Summary: Third-party question banks like UWorld, Blueprint, and Kaplan train students to recognize question types specific to that platform but not to understand the underlying concept. This forces a second round of adaptation when students finally switch to official AAMC materials.

DOK 1 Facts:

- UWorld, Blueprint, and Kaplan each have proprietary question styles, difficulty calibrations, and distractor logic that differ from AAMC's
- Students who over-rely on third-party QBanks consistently report “AAMC questions feel different” — the platform's patterns don't transfer
- Forces a mandatory de-adaptation phase before test day: time spent re-learning how the real test asks questions
- The dominant prep stack is AAMC + Anki + UWorld used together — no single platform captures a majority of students

[OBJ]

Category 2: Memory, performance, and readiness are three different things — every tool treats them as one

Sources: Roediger & Karpicke (2006, Psychological Science) · Kornell & Bjork (2008, Psychological Science) · Dunning & Kruger (1999, JPSP)

DOK 2 Summary: Recalling a flashcard, answering a practice question correctly, and being able to predict your score on test day are three distinct cognitive tasks — but every current MCAT tool collapses them into a single progress metric, producing confident-looking numbers that don't reflect actual exam readiness.

DOK 1 Facts:

- Anki's FSRS algorithm estimates probability of recall — not probability of getting a novel exam question right

- Retrieval practice group recalled 61% of material after one week vs. 40% for the restudy group — but recall of isolated facts doesn't predict passage-based reasoning performance
- Students systematically overestimate readiness when feedback comes from familiar question formats (Dunning-Kruger effect amplified by platform-specific patterns)
- A student can have high Anki retention on “the mitochondria is the powerhouse of the cell” and still fail an MCAT passage on cellular respiration — the bridge from memory to application is unbuilt
- No current tool shows: a score range, a confidence interval, what data is missing, or when it refuses to estimate

[OBJ]

Category 3: Students study in two contexts, but tools don't share one engine

Sources: Kornell (2009, Psychological Science) · Lindquist & Thorell (2019, Computers & Education)

DOK 2 Summary: Deep content work happens at a desk; retrieval practice happens on a phone between classes — but no current MCAT tool shares a single scheduling engine across both contexts, so reviews done on mobile don't inform the desktop's spaced repetition algorithm and vice versa, silently degrading retention.

DOK 1 Facts:

- Anki has separate desktop and mobile apps, but sync is manual and asynchronous — a review on AnkiMobile doesn't instantly update the desktop scheduler
- Blueprint, Kaplan, and UWorld are web-first; their mobile apps are companion viewers, not co-equal review engines
- When a student uses Anki for memory and UWorld for practice questions, there is no shared data layer — topic weakness signals from one tool never reach the other
- Students using fragmented stacks cannot reliably answer: “what is the single most valuable thing to study right now?”
- Distributed practice (studying across multiple short sessions) produces stronger retention than massed practice — but only if the scheduling algorithm has full context across all sessions

DOK 1

Study Tools:

- [UWorld](#)
 - Question bank with explanations, analytics, and timed/tutor modes
 - Paid: 300 to 1200
 - **Learning Science:** Retrieval Practice/Practice Based Learning
 - [Taking Memory Tests Improves Long-Term Retention](#)

- [Blueprint](#)
 - Adaptive study planner, full-length practice tests, 1600+ page online textbook, live/self-paced course options.
 - Paid: 99 to 3000 for personalized 1-on-1 tutoring
 - **Learning Science:** Spaced Repetition
 - [Science behind Spaced Repetition](#)
- [Kaplan MCAT](#)
 - 7 full-length practice exams, lectures, and a large question bank. Offers live, online self-paced, and tutoring options.
 - Paid: 1500-3600
 - **Learning Science:** No solid learning science
 - Structured curriculum and live instruction
 - [“As a result of this screening, 50 independent effects were identified that could be subjected to meta-analysis. The meta-analysis found that, on average, students in online learning conditions performed modestly better than those receiving face-to-face instruction”](#)
- [AAMC Materials](#)
 - Official practice exams, section banks, question packs. Made by the test-makers
 - **Learning Science:** No solid learning science
 - Real test simulation
 - Prepares students for what the real MCAT is like
 - AAMC writes both the real and practice exams, students can use the practice exams to gauge their level of understanding.
- [Anki \(+ Premed Decks\)](#)
 - Spaced-repetition flashcard app. Content memorization alongside a QBank.
 - Free to use with massive community-built decks
 - **Learning Science:** Spaced Repetition, Active Recall
 - [Anki is for remembering content—not learning new content.](#)

MCAT General Facts

- Total test-takers:
 - N = 305,494 total MCAT exams administered across the 2023, 2024, and 2025 testing years combined
 - roughly ~101,800 sittings per year across those three years

Tab 3

Owners

- Evan Cabrera

Purpose

- The purpose of this BrainLift is to redesign how MCAT students study by identifying the core failures of existing prep tools and building a desktop and mobile app, building off of Anki, that measures memory, performance, and readiness as three separate, honest scores backed by real learning science.

In Scope

- MCAT test-takers (pre-med undergraduates preparing for medical school admissions)
- Existing MCAT prep tools and their learning science foundations
- Spaced repetition and retrieval practice as retention mechanisms
- Score modeling: memory → performance → readiness
- Desktop and mobile study behavior patterns
- AI-generated flashcard quality and safety
- Anki's FSRS(Free Spaced Repetition Scheduler) algorithm and Rust engine

Out of Scope

- Other graduate exams (LSAT, GMAT, USMLE)
- General flashcard app design outside of MCAT context
- Medical school admissions strategy beyond the MCAT

DOK 4: Spiky Points of View (SPOVs)

- **Spiky POV:** Reading a wrong-answer explanation feels like learning, but it isn't. Instead of displaying the wrong-answer explanation, the question should instead be broken down into smaller questions to guide the user.
 - **Elaboration:** Every QBank delivers explanations as static text after a wrong answer. Students read them passively and believe they understand the answer. But passive re-reading is the least effective form of learning. Retrieval builds retention. A student who reads an explanation and doesn't retrieve the corrected concept within a scheduled window will fail that question again two weeks later. The fix isn't better explanations. It's forcing re-retrieval of the corrected concept on a spaced schedule, which no current MCAT tool does.
- **DOK 3: Insights**

- **Insight 1:** The illusion of understanding is seen in most MCAT prep guides.
 - Category 1 shows that QBanks teach platform patterns, not concepts. Category 2 shows that students confuse memory with readiness. Category 3 shows that reading explanations feels like learning but isn't. These prep guides cause students to constantly mistake a feeling of understanding for actual understanding.
- **Insight 2:** There is a gap between memory and actual performance/learning
 - Anki helps with building recall, but the MCAT tests application of recalled knowledge to passages under time pressure. Anki does not help in breaking down the passages, forcing students to spend more time adapting to the problems.

DOK 2: Knowledge Tree

- **Category 1: Question Banks build pattern recognition, not understanding**
 - **Subcategory 1.1: Third-Party QBanks****
 - Source: [UWorld](#), [Blueprint](#), [Kaplan MCAT](#)
 - **DOK 1 Facts:**
 - UWorld, Blueprint, and Kaplan each have their own question styles, and difficulty calibrations, that differ from AAMC's
 - Forces a mandatory de-adaptation phase before test day
 - There is no one size fits all platform for majority of students
 - **DOK 2 Summary:**
 - Third-party question banks train students to recognize question archetypes specific to that platform, not the underlying concept, creating a second adaptation problem when students switch to official AAMC materials
- **Category 2: Students mistake passive explanation reading for understanding**
 - **Subcategory 3.1: Passive Reading**
 - **DOK 1 Facts:**
 - [Retrieval practice group recalled 61% after one week; restudy group recalled*40%](#) -
 - Repeated restudying creates false confidence without improving long-term retention

- UWorld, Blueprint, and Kaplan all deliver wrong-answer explanations as static text
- A student can read an explanation, feel like they "get it," and fail the same concept two weeks later with no memory of the correction
- **DOK 2 Summary:**
 - When students get a question wrong and read the explanation, they convince themselves they understand the question, but passive re-reading makes false confidence without actual retention. Without forced re-retrieval on a spaced schedule, the correction doesn't stick.

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